MAKERERE UNIVERSITY

COLLEGE OF COMPUTING & INFORMATION SCIENCES

SCHOOL OF COMPUTING & INFORMATICS TECHNOLOGY

END OF SEMESTER I EXAMINATION 2020/2021

YEAR OF STUDY: II

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: BSE 2106

DATE: 22nd September 2021 TIME: 9AM - 6PM

EXAMINATION INSTRUCTIONS

a) **ATTEMPT ONLY FOUR QUESTIONS (25 MARKS EACH).**
b) **ANY PROOF OF PLAGIARISM (COPYING AND PASTING) FROM ANY SOURCE MAY LEAD TO CANCELLATION OF YOUR EXAM**

Question One

Suppose the network layer provides the following service. The network layer in the source host accepts a segment of maximum size 1,200 bytes and a destination host address from the transport layer. The network layer then guarantees to deliver the segment to the transport layer at the destination host. Suppose many network application processes can be running at the destination host.

a) Design the simplest possible transport-layer protocol that will get application data to the desired process at the destination host. Assume the operating system in the destination host has assigned a 4-byte port number to each running application process. **[10 marks]**
b) Modify this protocol so that it provides a "return address" to the destination process. **[5 marks]**
c) In your protocols, does the transport layer "have to do anything" in the core of the computer network? Explain your answer. **[4 marks]**
d) The Transmission Control Protocol (TCP) has got six flag fields. List these six fields and discuss the purpose of each of these fields. **[6 marks]**

Question Two

a) Explain the ARP protocol to a friend. In that explanation, mention what layer it operates at and the kind of service it provides to the layer above it. **[*6 Marks*]**
b) Having ARP table entries time out after 10 to 15 minutes is an attempt at a reasonable compromise. Describe the problems that can occur if the timeout value is too small or too large. **[*4 marks*]**
c) IEEE defines two sublayers at Layer 2, discuss the purpose of each. **[*4 Marks*]**
d) How does a computer attached to a shared LAN decide whether to accept a packet? **[*3 Marks*]**
e) Expand the acronym CSMA/CD, explaining each part and why CSMA/CD uses a random delay? **[*4 Marks*]**
f) Explain why collision is an issue in a random-access protocol but not in reservation or channel partitioning protocols. **[*4 Marks*]**

Question Three

a) Differentiate between:
   i. Routing and Forwarding. **[*3 Marks*]**
   ii. Manual IP address allocation and dynamic IP address allocation. **[*3 Marks*]**

b) Using an illustration, give the difference between packet switching and circuit switching. **[*4 Marks*]**

c) Discuss the following as they relate to IP networks:
   i. Network Address Translation. **[*3 marks*]**
   ii. Datagram Networks. **[*3 marks*]**
   iii. IP Fragmentation and Reassembly. **[*3 marks*]**
   iv. ICMP. **[*3 marks*]**
   v. ICANN. **[*3 marks*]**

Question Four

a) Discuss any two characteristics of IP Protocol. **[*4 Marks*]**

b) With explanation of the rules applied, provide the short form of the IPv6 address, 2041:0000:140F:079F:0000:0000:0000:131B? **[*4.5 Marks*]**

c) Suppose a router has built up the routing table shown in Table below:

SubnetNumber	SubnetMask	NextHop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.252.0	R3
(default)		R4

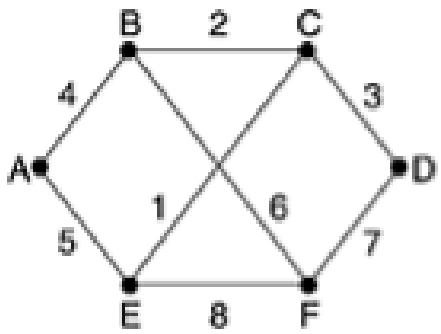
The router can deliver packets directly over interfaces 0 and 1, or it can forward packets to routers R2, R3, or R4. Assume the router does the longest prefix match. Describe what the router does with a packet addressed to each of the following destinations: **[*5 marks*]**

   i. 128.96.167.151.
   ii. 128.96.163.151
   iii. 128.96.169.192.

iv. 128.96.165.121.

v. 171.1.2.3

d) Consider the network shown on the link below, with the indicated link costs. Use Dijkstra's shortest path algorithm to compute the graph of the shortest path from A to all network nodes i.e. B, C, D, E, and F. **[*7.5 Marks*]**



e) Suppose that instead of using 16 bits for the network part of a class B address originally, 20 bits had been used. How many class B networks would there have been? **[*2 Marks*]**

f) A network on the Internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts it can handle? **[*2 Marks*]**

Question Five

Design and describe an application-level protocol to be used between an automatic teller machine (ATM) and a bank's centralized computer.

a) Your protocol should allow a user's card and password to be verified, the account balance (maintained at the centralized computer) to be queried, and an account withdrawal to be made. Your protocol entities should be able to handle the all-too-common case in which there is INSUFFICIENT BALANCE on the account to cover the withdrawal. Specify your protocol by listing the messages exchanged and the action taken by the ATM or the bank's centralized computer on transmission and receipt of messages. **[12 marks]**

b) Sketch the operation of your protocol for the case of a simple withdrawal with no errors, using a diagram similar to that in Figure 1 below. **[7 marks]**

c) Explicitly state the assumptions made by your protocol about the underlying end-to-end transport service. **[6 marks]**

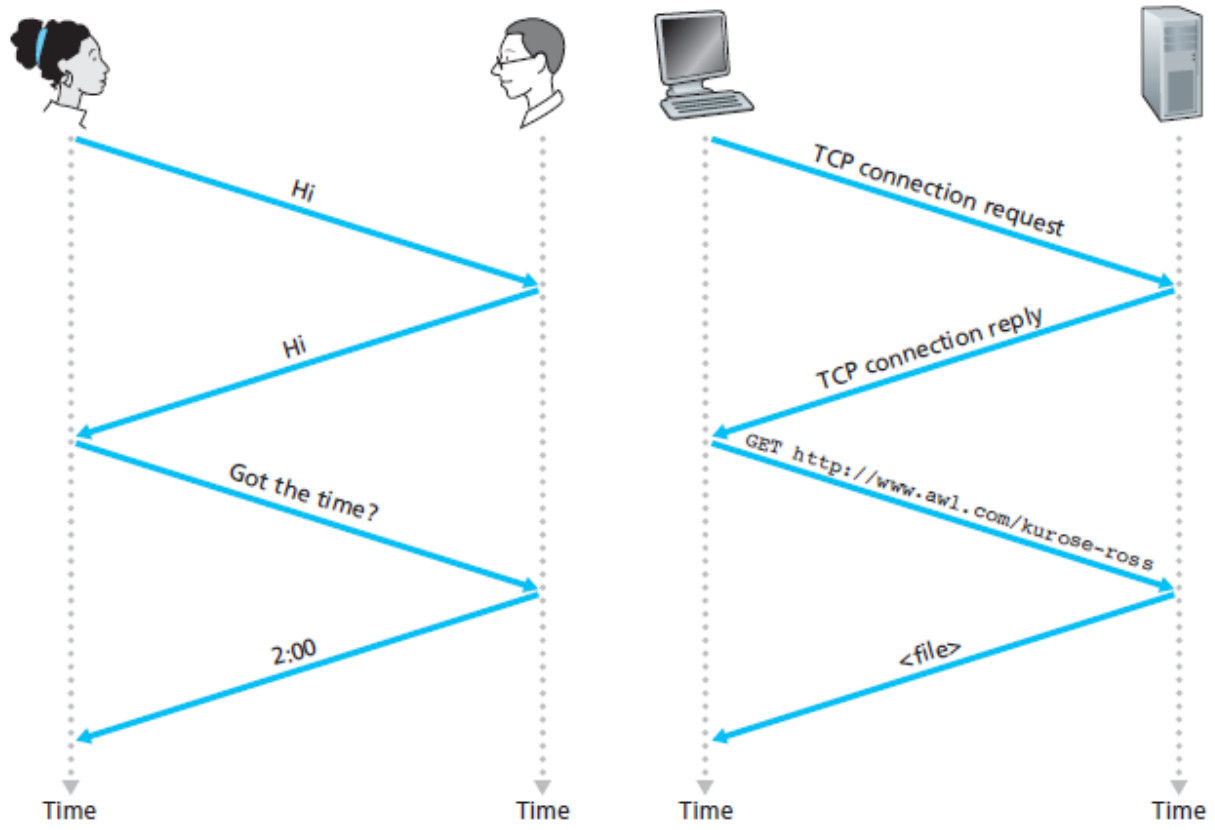


Figure 1: A human protocol and a computer network protocol

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